Structural Example

```
module majority (major, V1, V2, V3) ;

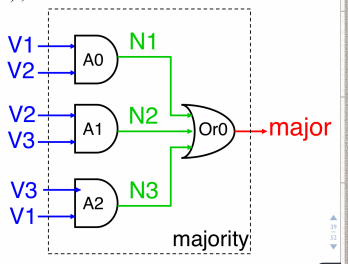
output major ;
input V1, V2, V3 ;

wire N1, N2, N3;

and A0 (N1, V1, V2),
    A1 (N2, V2, V3),
    A2 (N3, V3, V1);

or  Or0  (major, N1, N2, N3);

endmodule
```

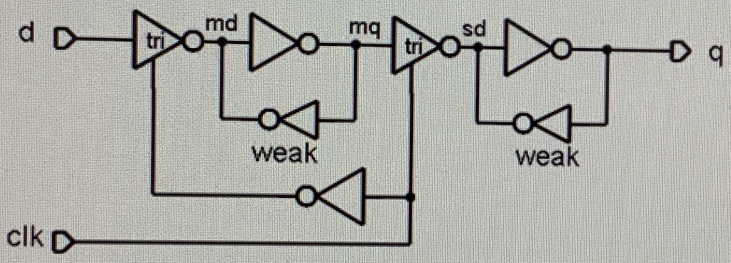


ECE 551 Exercise02

Master-slave latch (Flip flip)

This is the last problem of HW1 pos edge

Submit whatever you have for both the DUT and the testbench to the dropbox by end of class time.



a) Code this circuit structurally in Verilog (in Modelsim environment)(attach the Verilog code for this module)(**Note:** tri-state gates might need a 1 time unit delay modeled for proper simulation, an inverting tri-state is verilog primitive **notif1**).

b) b) Create a testbench, instantiate and simulate this circuit. Attach a print out of the waveforms, and the Verilog of the test bench

c) c) Add an active **high asynchronous** reset input to this circuit. Code this new circuit in Verilog and update your test bench to test the new input. Again attach Verilog and waveforms

d) Was the cost of adding the asynch reset high, medium, or low? Think in terms of number of transistors.

Test Bench

```
module mystery_tb();
  reg D_stim;
  reg clk;
  wire q_out;   // DUT will drive
d_stim, clk, q_out;
  // Instantiate DUT //
mystery iOUT(.clk(clk), .d(d_stim), .q(q_out)
initial begin
  clk = 0;
  d_stim = 0;
  #10;
  clk = 1;
```

Superset

System Verilog logic

generic units

```
@(posedge clk); #1;
if (q_out !== 1'b0) begin
```

#1;

// is q_out == 0

#10

clk=0;

Brute force

$display("ERR: q_out should be0");

$stop();

end

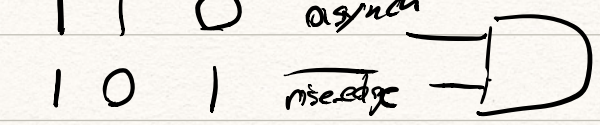
end

always

#5 clk = ~clk;

endmodule

asynch	nsedge	
1	1	0
1	0	1
0	1	0
0	0	0

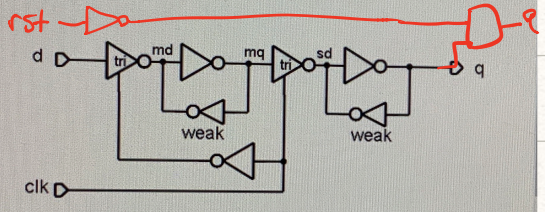


Asynch reset

ECE 551
Exercise02

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